

Week 6 – Advanced Testing & Debugging

Summary:

Week 6 was dedicated to **thorough testing of all backend functionalities** and debugging errors to ensure reliable operations across all supported languages.

Key Activities:

- Executed **test cases** for Python, C, C++, and Java programs, including edge cases.
- Verified **input/output handling**, compilation errors, and runtime exceptions.
- Debugged containerized environment issues (file permissions, temporary directories).
- Prepared **test reports** for each language.

Technologies Used: Python, FastAPI, Docker, GCC, g++, JDK, WSL2

Outcome:

Backend is fully tested and verified for **accuracy, stability, and error handling**. All issues resolved, ensuring reliable multi-language execution.